

Supplementary Material

Real-World Industrial Process Monitoring Benchmark for Bio-feedstock Co-processing in Fluid Catalytic Cracking Units

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This document reports the analyses referred to in our response to Reviewer 1. Section S1 describes the released data files and checks their consistency. Section S2 reports the error analysis of the reference model requested under Comment 3. Section S3 reports the quantitative analysis of renewable LCC requested under Comment 5. Section S4 lists the evaluated baselines that are excluded from the leaderboard. All numbers are produced by analysis/round3_supplementary_analysis.py, and the corresponding result tables are stored in the results folder of the repository <https://github.com/Liangcao2018/CEP-FCC-Benchmark>. Sections S1 and S2 and Table S8 can be regenerated from the released files alone. The benchmark-window results of Section S3 require plant values in physical units, which are confidential, and are therefore reported in dimensionless form.

S1. Released data and reproducibility

The absolute operating values of the unit, such as feed and product rates and operating temperatures, are commercially sensitive. Because the normalized benchmark series is public, releasing the true normalization parameters would be equivalent to releasing those values. Every variable, including the target, is therefore released on a masked scale. The masked value of each variable is a positive linear transformation of its engineering-scale value, $x' = s \cdot x + o$ with $s > 0$, and the coefficients s and o differ between variables and are not disclosed.

A positive linear transformation leaves the z-score of every sample unchanged. Applying (masked value minus Mean) divided by Std with the values in normalization_parameters.csv therefore reproduces fcc_normalized_data.csv to within 2×10^{-12} for every variable. The benchmark code, code.py, reads the masked file and applies the z-score transform itself, and it reproduces the test R^2 of the ten deterministic linear baselines in Table 6 of the manuscript to four decimal places. The masked file contains the same 104,000 samples as the normalized file, in chronological order and at a fixed spacing of 10 minutes, with a sequential index in place of timestamps. Table S1 lists the released files.

Table S1. Files released in the repository.

File	Content
fcc_normalized_data.csv.zip	z-scored benchmark series, 104,000 samples, target and 15 features
data/fcc_masked_data.csv.zip	the same samples before z-scoring, on a masked scale
data/normalization_parameters.csv	mean and standard deviation of each masked variable, with physical quantity and recording unit
code.py	benchmark code; reads the masked file and performs the normalization
results/benchmark_results_27_models.csv	train and test metrics of the 27 reported baselines
results/excluded_models_negative_test_R2.csv	the three evaluated baselines with negative test R^2
results/best_predictions_test.csv	Transformer predictions on the test set (normalized scale)
analysis/round3_supplementary_analysis.py	script that produces Sections S1 to S3

Table S2 gives the physical quantity of each variable and the unit in which the plant records it. The units were determined from the engineering-scale records and from the physical relations that link the variables. For example, the recorded catalyst-to-oil ratio, catalyst circulation rate, and feed flows are mutually consistent only when the circulation rate is expressed in short tons per minute.

Table S2. Benchmark variables and recording units. The released values are on a masked scale.

Variable	Physical quantity	Recording unit
Target	Total LCC production	bbl/day
Feature_01	Fossil feed flow	bbl/day
Feature_02	Bio-feed flow	bbl/day
Feature_03	Outlet flow 1	kbbbl/day
Feature_04	Outlet flow 2	kbbbl/day
Feature_05	Catalyst cooler fluffing air	kSCF/h
Feature_06	Catalyst circulation rate	short ton/min
Feature_07	Flue gas excess oxygen	vol%
Feature_08	CO concentration signal	ppm
Feature_09	Riser temperature	°F
Feature_10	Preheat temperature	°F
Feature_11	Catalyst cooler steam flow	klb/h
Feature_12	Upper regenerator temperature	°F
Feature_13	Catalyst to oil ratio	dimensionless
Feature_14	Reactor temperature	°F
Feature_15	Conversion rate during co-processing	%

The normalization parameters are computed over the full series, which is how the released benchmark code applies the z-score transform. Because statistics taken over the full series include the test period, we checked their influence by refitting the linear baselines with statistics taken from the training period only. Table S3 shows that the test R^2 changes by at most 0.0009, so the reported results do not depend on this choice. This check is also unaffected by the masking.

Table S3. Test R^2 of the linear baselines under full-series and training-only normalization.

Model	Full series	Training only	Difference
ElasticNet	0.4590	0.4590	+0.0000
Lasso	0.4519	0.4528	+0.0009
OLS	0.4434	0.4434	-0.0000
Ridge	0.4436	0.4436	-0.0000

S2. Error analysis of the reference model (Comment 3)

All results in this section refer to the Transformer predictions on the chronological test set of 10,399 samples. They are expressed on the normalized scale, in units of the full-series standard deviation of the target (σ), so that they can be regenerated directly from the released files. A positive error means that the model under-predicts LCC production.

Table S4. Overall test performance on the normalized scale.

Metric	Value
R^2	0.5376
RMSE (σ), i.e. N-RMSE	0.5295
MAE (σ)	0.4258
Mean error, actual minus predicted (σ)	0.2828

The mean squared error was decomposed into a bias part, a variance part, and an unsystematic part following Theil. Table S5 shows that 28.5% of the error comes from a constant offset and 10.8% from the predictions being less dispersed than the observations (standard deviation 0.604σ against 0.779σ over the test set). The remaining 60.6% is unsystematic.

Table S5. Decomposition of the test mean squared error.

Component	Share of MSE
Bias (constant offset)	28.5%
Variance (under-dispersion)	10.8%
Covariance (unsystematic)	60.6%

To examine heteroscedasticity, the test samples were divided into quintiles of the actual LCC production. Table S6 and Figure S1 show that the error depends strongly on the level of production. In the lowest quintile the model over-predicts by 0.19σ on average, whereas in the highest quintile it under-predicts by 0.66σ . The RMSE rises from 0.35σ to 0.76σ . Together with the variance component in Table S5, this indicates that the model compresses the range of LCC production towards its mean.

Table S6. Test error by quintile of actual LCC production (normalized scale).

Bin	Range (σ)	n	Mean error (σ)	RMSE (σ)
Q1	-2.26 to -0.44	2080	-0.185	0.347
Q2	-0.44 to 0.14	2080	0.171	0.357
Q3	0.14 to 0.54	2079	0.348	0.497
Q4	0.54 to 0.91	2080	0.422	0.567
Q5	0.91 to 1.91	2080	0.659	0.765

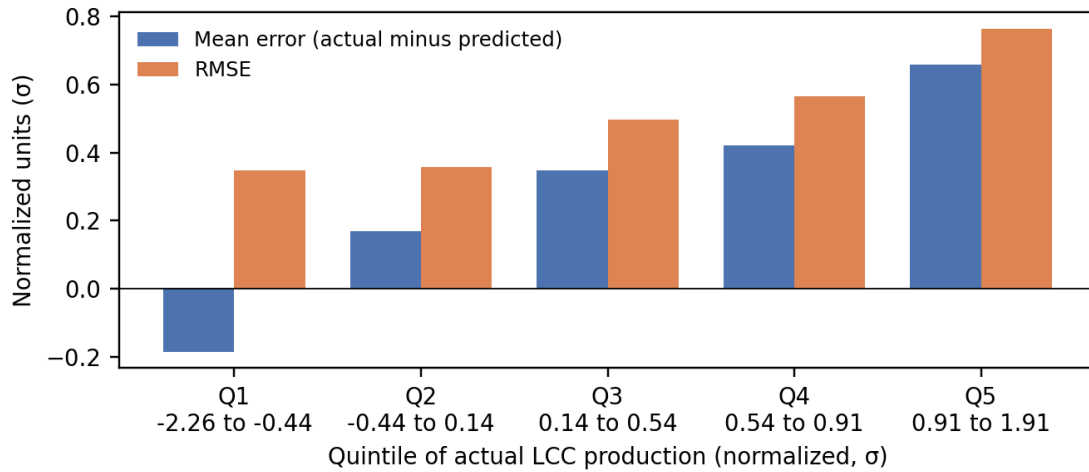


Figure S1. Mean error and RMSE of the Transformer by quintile of actual LCC production on the test set, in units of the target standard deviation.

Table S7 reports the effect of removing the constant offset of 0.283σ , which raises R^2 from 0.5376 to 0.6695. The residuals are strongly autocorrelated, with $ACF(1) = 0.998$. For this reason the confidence interval of the bias-corrected R^2 is computed with a moving-block bootstrap, which resamples contiguous blocks and therefore respects the dependence between neighbouring samples. The resulting intervals are wide, which supersedes the narrower interval quoted in an earlier response that assumed independent samples. We report the recalibration result on its own terms and draw no conclusion about a performance ceiling from it.

Table S7. Constant offset, bias-corrected R^2 , and residual autocorrelation.

Quantity	Value
R^2 before correction	0.5376
Constant offset (σ)	0.283
R^2 after removing the offset	0.6695
Residual $ACF(1)$	0.998
95% CI, block bootstrap, 1-day blocks	[0.552, 0.774]
95% CI, block bootstrap, 7-day blocks	[0.464, 0.837]

S3. Renewable LCC and the LCC production target (Comment 5)

The renewable portion of LCC is obtained with the model of Cao et al. (2024), which was estimated on the same 104,000-sample dataset. Total LCC production is written as $LCC = a \cdot F + b \cdot B + \epsilon$, where F and B are the fossil and bio feed flows, $a = 0.542$ and $b = 0.323$ are the yields per unit volume of each feed, and ϵ collects the part of LCC not explained by the two feeds. With the co-processing ratio $r = B/(F + B)$, renewable LCC is $b \cdot B + r \cdot \epsilon$, and the renewable fraction is $f = \text{renewable LCC} / LCC$ (Eqs. 8 and 9 in Cao et al., 2024).

This model needs the fossil feed, the bio feed, and LCC production in physical units, which the masked release does not provide. The benchmark-window results in Tables S9 and S10 and Figure S2 were therefore computed by the authors with the released script from the engineering-scale records (option `--engineering-data`), and they are reported in dimensionless form. Table S8 uses published values only and can be reproduced from the repository.

The fraction model is checked against the four ^{14}C measurements reported by Cao et al. (2024), listed in Table S8. The model reproduces the measured fraction with a mean absolute error of 0.27 percentage points and a mean bias of +0.21 percentage points, or 3.9% in relative terms. A linear fit of the measured fraction on the co-processing ratio gives a slope of 0.81, an intercept of -2.39 percentage points, and $R^2 = 0.92$. With four samples this fit is indicative only. These measurements were taken before the benchmark period, so no ^{14}C sample falls inside the 104,000-sample window.

Table S8. Measured and modelled renewable fraction of LCC (^{14}C data from Cao et al., 2024, Table 3).

Sample	Co-processing ratio (%)	Measured ^{14}C fraction (%)	Model fraction (%)	Error (pp)
1	10.75	6.30	6.68	+0.38
2	12.24	7.30	7.65	+0.35
3	12.32	7.80	7.69	-0.11
4	12.22	7.40	7.64	+0.24

Applying the model over the benchmark window answers the question of whether the relationship between LCC production and renewable carbon is linear or process-dependent. Table S9 and Figure S2 show that it is process-dependent. The co-processing ratio averages 9.77% with a standard deviation of 6.63%, and it is below 1% for 19.3% of the time. As a result, the renewable fraction varies widely, with a mean of 6.53% and a coefficient of variation of 68.9% (44.2% when co-processing is active). The fraction is almost linear in the co-processing ratio (slope 0.678, $R^2 = 0.997$). By contrast, when renewable LCC and total LCC are both expressed relative to their means over the window, so that strict proportionality would give a slope of 1 and an intercept of 0, the regression of renewable LCC on total LCC gives a slope of 1.80, an intercept of -0.80 , and $R^2 = 0.052$. On a logarithmic scale, 94.1% of the variance of renewable LCC comes from the fraction and 1.4% from total LCC production.

Table S9. Renewable fraction and renewable LCC over the benchmark window (104,000 samples).

Quantity	Value
Co-processing ratio, mean (SD), %	9.77 (6.63)
Co-processing ratio, maximum, %	22.88
Share of time with ratio below 1%	19.3%
Renewable fraction f , mean (SD), %	6.53 (4.50)
Renewable fraction f , coefficient of variation	68.9%
Renewable fraction f , mean and CV when co-processing	8.07%, 44.2%
Renewable LCC, coefficient of variation	70.2%
Renewable LCC on total LCC, both relative to their means: slope, intercept, R^2	1.80, -0.80 , 0.052
Fraction f on co-processing ratio: slope, intercept (pp), R^2	0.678, -0.094 , 0.997
Variance of log renewable LCC: fraction, total LCC, covariance	94.1%, 1.4%, 4.4%

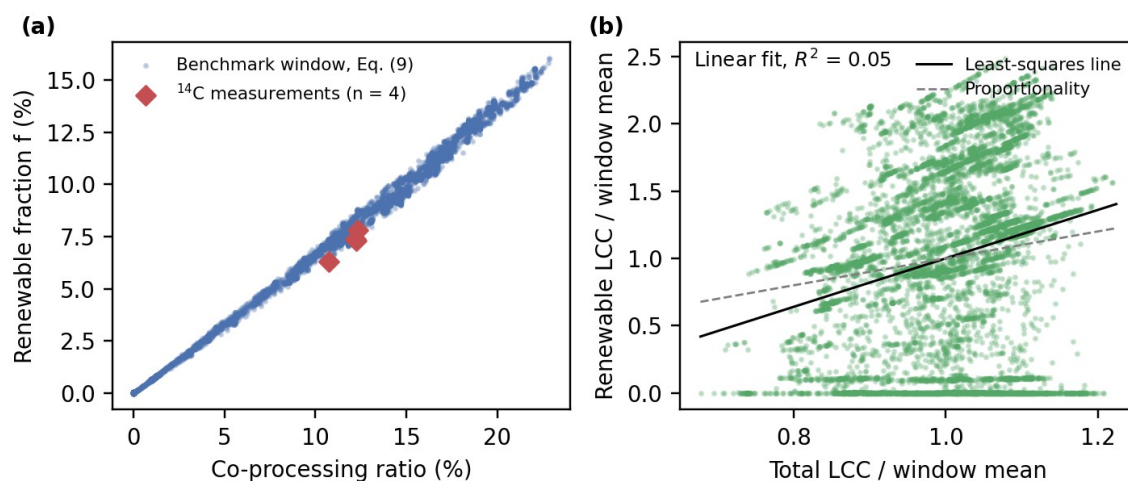


Figure S2. (a) Renewable fraction against co-processing ratio over the benchmark window (random subsample of 8,000 points) with the four ^{14}C measurements. (b) Renewable LCC against total LCC production, both relative to their means over the window, with the least-squares line and the line of strict proportionality.

Table S10 reports two renewable-carbon estimators on the benchmark test set. The two-stage estimator replaces measured LCC in Eq. (8) with the Transformer prediction from the main benchmark task, and the constant-fraction estimator multiplies measured LCC by the average fraction of the training period. The reference is Eq. (8) evaluated with measured LCC, so these figures isolate the error that the LCC soft sensor passes on to renewable LCC. The error of the fraction model itself is the one characterized against ^{14}C in Table S8. The two-stage estimator reaches $R^2 = 0.986$ with an RMSE of 8.2% of the mean renewable LCC over the test set, whereas the constant-fraction estimator reaches only $R^2 = 0.011$. A regressor trained directly on ^{14}C labels is not reported, because four labels cannot support separate training and testing.

Table S10. Renewable-carbon estimators on the benchmark test set (reference: Eq. 8 with measured LCC). Errors are given as a percentage of the mean reference value.

Estimator	R^2	RMSE (%)	MAE (%)	Mean error (%)
Two-stage: Transformer LCC + Eq. (8)	0.986	8.2	5.4	4.4
Constant fraction (6.14%) $\times$ measured LCC	0.011	69.3	62.8	13.7

S4. Baselines excluded from the leaderboard

The benchmark code evaluates 30 models. The three models in Table S11 have a negative test R^2 , meaning that they perform worse than predicting the mean, and they are therefore not included among the 27 reported baselines. Their metrics are released for completeness.

Table S11. Evaluated baselines with negative test R².

Model	Train R ²	Test R ²	Test N-RMSE
PassiveAggressive	0.6062	−0.4026	0.9221
RANSAC	−4.7439	−9.8312	2.5624
Lars	−1.329e+07	−3.335e+07	4.496e+03

Reference

Cao L., Su J., Saddler J., Cao Y., Wang Y., Lee G., Siang L.C., Pinchuk R., Li J., Gopaluni R.B. Real-time tracking of renewable carbon content with AI-aided approaches during co-processing of biofeedstocks. *Applied Energy* 2024;360:122815.